FEDERAL SPECIFICATION

LIGHT, COLOR PERCEPTION TESTING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification describes a color vision test lamp with daylight filter, to be used in conjunction with pseudoisochromatic plates for testing human color vision.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU–P–271 — Paper, Wrapping, Waterproofed Kraft.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipments (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.
MIL–C–11456 — Crates; Wood, Nailed, Unsheathed; Domestic Shipment, 1500 Pounds Maximum Load.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.**—The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Underwriters' Laboratories, Inc.:

Underwriters' Laboratories, Inc. Standards.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 161 Sixth Ave., New York 13, N. Y., 207 East Ohio Street, Chicago 11, Ill., and 1655 Scott Blvd., Santa Clara, Calif.)

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American Society for Testing Materials Publication:

ASTM–D–1684–59T—Tentative Recommended Practice for Lighting Cotton Classing Rooms for Color Grading.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 8, Pa.)

American Iron and Steel Institute:

Steel Products Manual.
Stainless and Heat Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each bidder shall submit to the contracting agency, proof that the electrical components he proposes to supply under this specification conform to the requirements of Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the electrical components conform to this requirement.

3.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his electrical components conform to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.2 Material

3.2.1 *Glass.* — Glass used in the fabrication of color filter shall be free from cracks, well annealed, and free of strain.

3.2.2 *Uprights.* — Tubing used in the fabrication of uprights shall be American Iron and Steel Institute (AISI) 302 or (AISI) 304 stainless steel.

3.2.3 *Base and bookrest.* — Base and bookrest shall be fabricated of cast iron.

3.2.4 *Luminaire.* — The luminaire shall be fabricated of sheet steel or aluminum.

3.3 Style, design and dimensions. — Shall be an easel lamp for holding and illuminating pseudoisochromatic plates, consisting essentially of a combination base and book holder, uprights, bookrest, luminaire with switch, incandescent lamp, color filter, and a length of electrical connection cord. The unit shall be in accordance with figure 1, and as hereinafter specified.

3.3.1 *Base and bookrest.* — The base and bookrest shall be so designed as to be suitable for properly supporting color test plates in multi-ring looseleaf binders. The base shall be provided with a recess deep enough to retain the board covers, and an opening shall be provided at the center of the base and bookrest as a relief for the back binding of these color vision test sets.

3.3.2 *Luminaire.* — The luminaire shall be designed to accommodate a color filter that shall be replaceable in case of breakage and removable for lamp replacement. The incandescent source shall be so shielded by the luminaire and filter that no light unmodified by the filter can escape. The luminaire shall be fitted with a lampholder (socket) designed for use with a medium screw base 100-watt lamp, and with a switch for turning the lamp on and off. The switch may be integral with the lampholder. The luminaire shall be provided with a 100-watt, shape A–21, 110-120 volt, inside frosted, incandescent lamp, $5\frac{5}{16}$ inches overall, with medium screw base.

3.3.3 *Color filter.* — The color filter shall be fabricated of glass, and shall filter the incandescent lamp to a color no yellower than a color temperature of 6500°Kelvin (K) and no bluer than a color temperature of 7500°K. In addition, the filter shall be such as to modify the energy distribution of the incandescent lamp (see 3.3.2) to approximate the energy distribution of International Commission on Illumination (C.I.E.) illuminant "C". The approximation shall be equal to, or better than, that produced by the incandescent lamp combined with a filter of Corning Daylite glass No. 5900. The degree of approximation shall be assessed by means of the "conformity index" specified in American Society for Testing Material (ASTM) Method

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3.3.4 *Thermal shock.* — The tubes shall ... The lower and upper tubes shall be ...

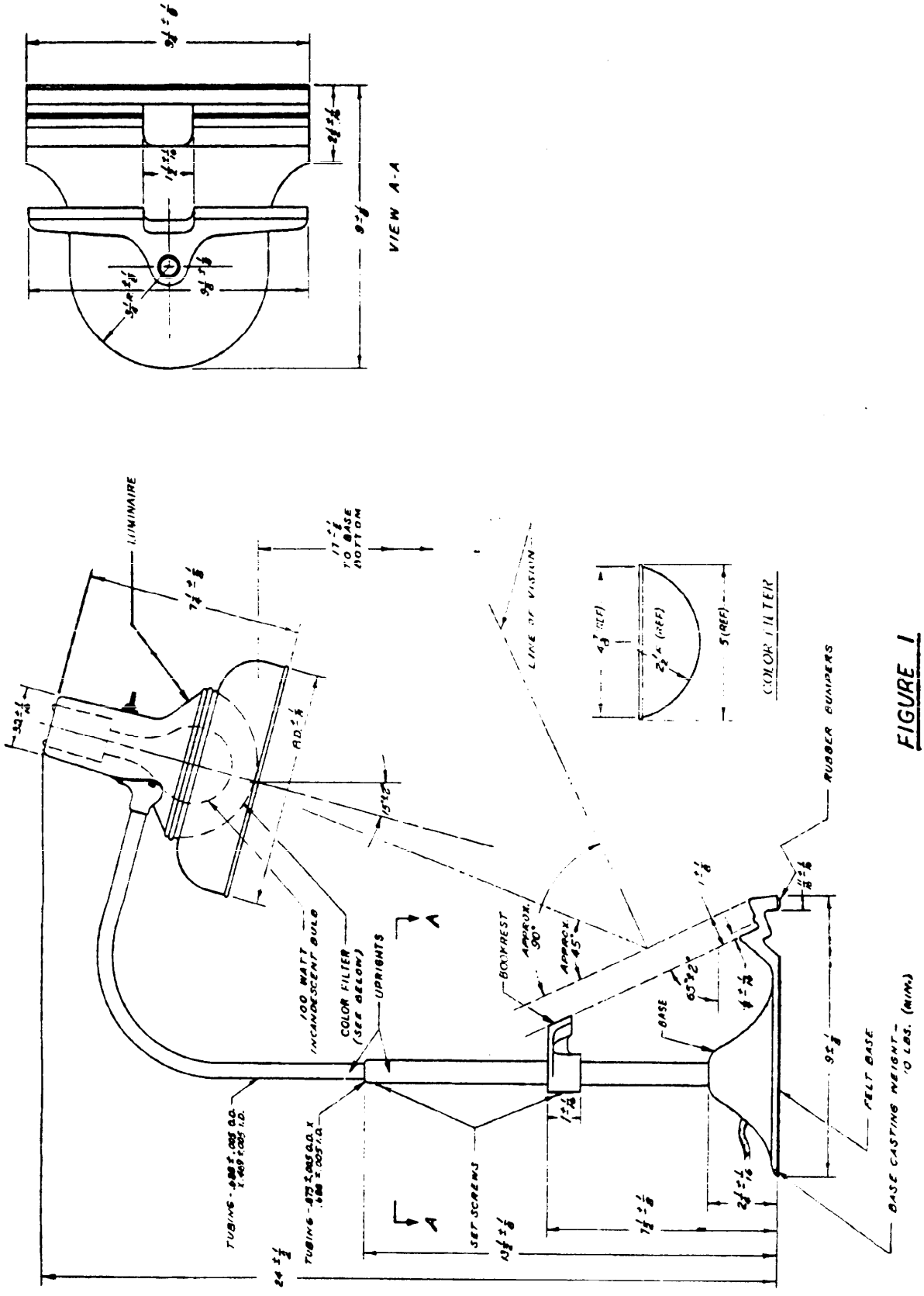


FIGURE 1

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D1684–59T; that is the luminaire shall have a conformity index to C.I.E. illuminant "C" not larger than that produced by combining an incandescent lamp with a thickness of Corning Daylite glass No. 5900, such as to produce a correlated color temperature as specified above.

3.3.4 *Electrical connection wire.*—A six to eight foot electrical connection wire, fitted at one end with an attachment plug shall be furnished with each unit.

3.4 **Finish.**—The base, upright, and exterior surfaces of the luminaire shall be black wrinkle finished. The interior surfaces of the luminaire shall be finished in such a manner to produce an efficient reflecting surface.

3.5 **Performance.**—When measured as specified in 4.4.1, in the plane of the open color vision test plate book, each point, four inches from the center of the book, shall receive not less than twenty nor more than thirty foot-candles of illumination.

3.6 **Identification marking.** — The light shall be clearly and legibly marked with the name or registered trademark of the manufacturer.

3.7 **Workmanship.**—Workmanship shall be first class throughout. Lights shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.** —The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* — When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**

4.2.1 *For examination.* — Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

4.2.2 *For tests.* — Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105. Requirement applicable to individual unit only.

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL
Light (end item)	Performance	3.5	4.4.1	1-light	L8*	*

* Acceptance number shall be zero for all sample sizes.

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4.2.3 *Raw materials.* — Four ounces of each lot of each metal used in the construction of the uprights shall be selected for chemical analysis.

4.2.4 *Components.* — Sufficient samples of the components shall be selected from each lot of material in the manufacture of the lights to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than two samples shall be taken regardless of lot size.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination.—The lights shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. — Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Performance test.*—Illumination shall be measured with a photoelectric photometer.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging, and packing protection for supplies and equipment.

5.1 Packaging.

5.1.1 *Unit of issue.* — One (1) light, as specified, constitutes one unit.

5.1.2 *Level B.*

5.1.2.1 *Unit package.* — Each unit shall be disassembled as far as is practicable. Accessories and component parts shall be packaged in one or more fiberboard cartons or nailed wood boxes of suitable size and design. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.1.3 *Level A.*

5.1.3.1 *Unit package.* — Each unit shall be packaged as specified for level B. In addition, reflector and electrical components shall be further packaged in accordance with Method IA–2 or IA–8 of Military Specification MIL–P–116.

5.1.4 *Level C.* — Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* — Each unit shall be overpacked in an exterior container designed for type 3 load and constructed in accordance with Federal Specification PPP–B–601, table II, using type III, class 1 plywood or Federal Specification PPP–B–621, table IV. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces. All box panels, except base, shall be lined with paper conforming to the requirements of Federal Specification UU–P–271, class H–4, L–2(a), L–2(b), or M–3, in such manner as to prevent direct entry of water. Closure shall comply with appendix of applicable box specification.

5.2.1.2 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.2 *Level B.*

5.2.2.1 *Exterior container.* — Each unit shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP–B–601, table I, using type III, class 1 plywood; Federal Specification PPP–B–621, table II; or Military Specification MIL–C–11456. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Nonmilitary.* — In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military.*

5.3.2.1 *Interior packages.*—Each container shall be marked to identify contents.

5.3.2.2 *Exterior container.* — Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B preservation and packing is intended to provide economical but limited protection. and should be specified only when it is determined the lights will be held in covered storage approximately 12 months from date of initial packaging.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 **Transportation description.**—Transportation description applicable to this item is:

Lamps electric, with globe or shades not exceeding 21 inches in diameter.
Carload minimum weight 14,000 pounds subject to Rule 34 of the Uniform Freight Classification.
Truckload minimum weight 14,000 pounds per National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.

Copies of this specification may be purchased for 5 cents each.

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